Cloud Microphysics Value Added Product
PRELIMINARY Version
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As this is a preliminary dataset, we invite users to use the following contact point to offer feedback, corrections, and questions:
Elise Rosky (emrosky@ucar.edu)

This preliminary dataset contains a best estimate of cloud microphysical parameters from CAESAR. The dataset is constructed using a combination of all the available in-situ cloud probes on board the C-130 aircraft. By analyzing the agreement between probes, and selecting appropriate probes for specific particle size ranges, we obtain particle concentration estimates across the full particle size spectrum from 2um to 19mm, and PRELIMINARY uncertainty bounds on these data.

The variables are as follows:

```
dimensions:
  time = [varies per flight]
  bin_centers = 169
  bin_edges = 170
variables:
  double time(time) ;
    time:longname = "UTC time for concentration arrays and bulk variables" ;
    time:units = "Seconds from midnight of start date" ;
  double bin_edges(bin_edges) ;
    bin_edges:longname = "Upper/lower edges of concentration size bins" ;
    bin_edges:units = "microns" ;
  double bin_centers(bin_centers) ;
    bin_centers:longname = "Center value of concentration size bins" ;
    bin_centers:units = "microns" ;
  int cloud_phase(time) ;
    cloud_phase:longname = "Estimate of cloud phase identification using algorithm" ;
    cloud_phase:units = "none" ;
    cloud_phase:flag_values = 0, 1, 2, 3, 4 ;
    string cloud_phase:flag_meaning = "clear", "ice", "mixed", "liquid", "drizzle" ;
  double concentration(bin_centers, time) ;
    concentration:longname = "Best estimate of particle number concentration, all particles, normalized by bin width" ;
    concentration:units = "#/m4" ;
  double concentration_err(bin_centers, time) ;
    concentration_err:longname = "Uncertainty bound on particle number concentration, all particles, normalized by bin width" ;
    concentration_err:units = "#/m4" ;
  double concentration_round(bin_centers, time) ;
    concentration_round:longname = "Best estimate of particle number concentration, round particles, normalized by bin width" ;
    concentration_round:units = "#/m4" ;
  double concentration_ice(bin_centers, time) ;
    concentration_ice:longname = "Best estimate of particle number concentration, ice particles, normalized by bin width" ;
    concentration_ice:units = "#/m4" ;
  double concentration_round_err(bin_centers, time) ;
    concentration_round_err:longname = "Uncertainty bound on particle number concentration, round particles, normalized by
bin width" ;
    concentration_round_err:units = "#/m4" ;
```

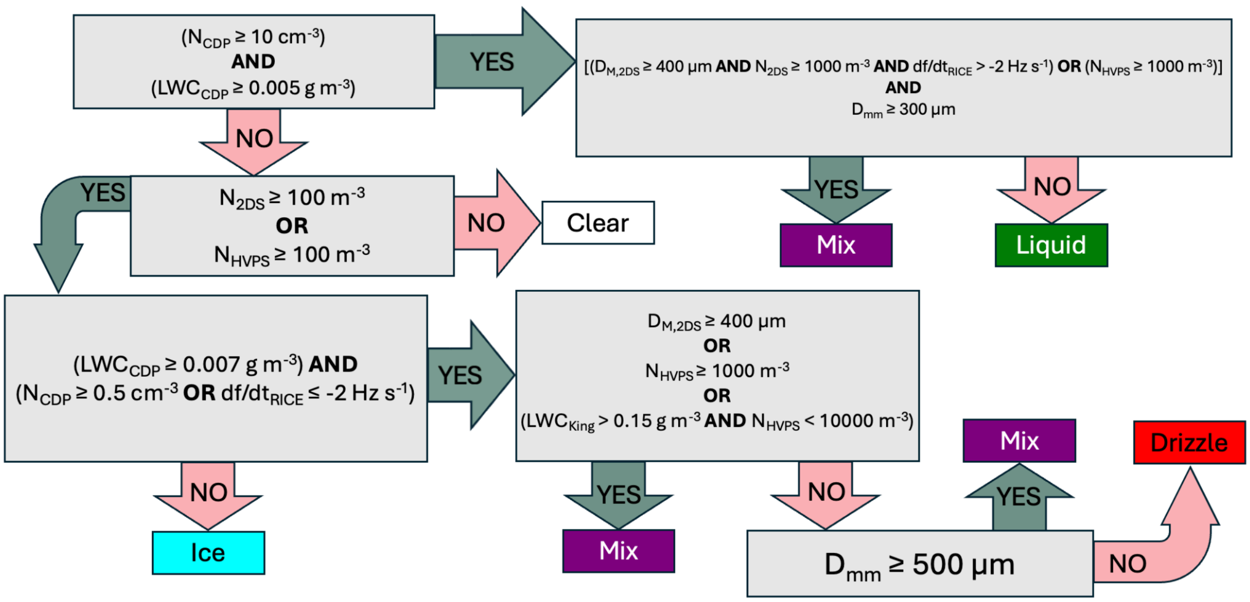
```
    double concentration_ice_err(bin_centers, time) ;
        concentration_ice_err:longname = "Uncertainty bound on particle number concentration, ice particles, normalized by bin width" ;
        concentration_ice_err:units = "#/m4" ;
    double lwc(time) ;
        lwc:longname = "Best estimate of liquid water content from Nevzorov" ;
        lwc:units = "g/m3" ;
    double twc(time) ;
        twc:longname = "Best estimate of total water content from Nevzorov" ;
        twc:units = "g/m3" ;
    double iwc(time) ;
        iwc:longname = "Best estimate of ice water content from Nevzorov" ;
        iwc:units = "g/m3" ;
    double lwc_err(time) ;
        lwc_err:longname = "Uncertainty bound on liquid water content" ;
        lwc_err:units = "g/m3" ;
    double twc_err(time) ;
        twc_err:longname = "Uncertainty bound on total water content" ;
        twc_err:units = "g/m3" ;
    double iwc_err(time) ;
        iwc_err:longname = "Uncertainty bound on ice water content" ;
        iwc_err:units = "g/m3" ;
    double lat(time) ;
        lat:longname = "Latitude" ;
        lat:units = "degrees North" ;
    double lon(time) ;
        lon:longname = "Longitude" ;
        lon:units = "degrees East" ;
    double alt(time) ;
        alt:longname = "GPS Altitude" ;
        alt:units = "meters" ;
    double t(time) ;
        t:longname = "Ambient Temperature" ;
        t:units = "C" ;
    int concentration_probe(bin_centers, time) ;
        concentration_probe:longname = "Identification of probe(s) used to construct the concentration best estimate" ;
        concentration_probe:units = "none" ;
        concentration_probe:flag_values = 1, 2, 3, 4, 5, 6, 7 ;
        string concentration_probe:flag_meaning = "CDP", "CDP_HOLODEC_mean", "HOLODEC", "HOLODEC_F2DS_mean", "F2DS", "F2DS_HVPS_mean", "HVPS" ;
    int concentration_round_probe(bin_centers, time) ;
        concentration_round_probe:longname = "Identification of probe(s) used to construct the concentration best estimate" ;
        concentration_round_probe:units = "none" ;
        concentration_round_probe:flag_values = 1, 2, 3, 4, 5, 6, 7 ;
        string concentration_round_probe:flag_meaning = "CDP", "CDP_HOLODEC_mean", "HOLODEC", "HOLODEC_F2DS_mean", "F2DS", "F2DS_HVPS_mean", "HVPS" ;
    int concentration_ice_probe(bin_centers, time) ;
        concentration_ice_probe:longname = "Identification of probe(s) used to construct the concentration best estimate" ;
        concentration_ice_probe:units = "none" ;
        concentration_ice_probe:flag_values = 1, 2, 3, 4, 5, 6, 7 ;
        string concentration_ice_probe:flag_meaning = "CDP", "CDP_HOLODEC_mean", "HOLODEC", "HOLODEC_F2DS_mean", "F2DS", "F2DS_HVPS_mean", "HVPS" ;
```

The methods of constructing each variable are documented below.

The code used to produce the dataset is publicly available in the following git repository, the documentation below refers to file names in this repository with italicised font.
https://github.com/erosky/CAESAR-microphysics/

CLOUD PHASE ID:

The cloud phase identification is preliminary, produced by The Cloud Physics Group at CIWRO, University of Oklahoma. The cloud phase data and flowchart shown below was provided by Nick Amundsen. The current contact person for the phase identification is Greg McFarquhar (mcfarq@ou.edu).



BULK LWC, TWC, IWC:

The Nevzorov probe is used for a best estimate of LWC and TWC for all flights. The Nevzorov data was quality controlled by Shane Maritch and Jeff French.

The original Nevzorov data is available in the CAESAR archive. For this value-added dataset we apply the following operations (*read_nevzorov.m*):

- NaN values are inserted at times where Nevzorov data is flagged as "conditional" or "bad".
- Negative LWC and TWC values are set to zero.
- IWC is estimated by taking the difference between LWC and TWC.

Uncertainty bounds:

- Baseline LWC and TWC uncertainty bounds are assigned as 0.01 or 0.02 g/m3 depending on the Nevzorov quality flag.
- By comparing the Nevzorov LWC with the King probe LWC across all flights, we assign an additional 10% uncertainty bound to the Nevzorov values.
- The assigned uncertainty is the greater of the baseline uncertainty and the 10% bound.
- IWC uncertainty is assigned through propagation of error from the LWC and TWC values.

PARTICLE NUMBER CONCENTRATION:

(*PSD_merge.m*) The rules for combining multiple cloud probe data into one particle size distribution are applied at each timestep. Identification of which probe was selected in each instance is available in the "concentration_probe" variable.

(*create_bins.m*) BIN EDGE DEFINITIONS:

```
probe_endbins.fixed_cdp =  [1.93  2.91  3.89  4.87  5.85  6.83  7.81  8.79  9.77 ...
   10.75 11.73 12.71];

probe_endbins.crossover_cdp_holo = [12.71 13.69 15.65 17.61];

probe_endbins.fixed_holo = [17.61 19.57 21.53 23.49 25.45 27.41 29.37 31.34 33.3  35.26 ...
   37.22 39.18 41.14 43.1  45.0];
```

- CDP used as backup if HOLODEC unavailable

```
probe_endbins.crossover_holo_f2ds = [ 45.  55.   65.   75.   85.   95. 105.  125.  ...
   145.  175.  225.];

probe_endbins.fixed_f2ds = [225. 275.  325.  400.  475.  550.  625.  700. 825.];

probe_endbins.crossover_f2ds_hvps = [825.  975.  1125.   1275.   1425.];

probe_endbins.fixed_hvps = [1425… 19275];
```

Fixed size ranges are size bins that use the data from a single assigned probe.

Crossover regions are size bins that will use a combination of values from two probes that overlap in that size range.

(*transition.m*) If both probes have non-zero values for over half of the crossover region:

> A mean of the two probes' values is taken when both probes are non-zero. If one probe=0 but the other has a value, the nonzero value is used. If both probes =0, then the value is zero.
>
> This preference for a nonzero value is used because probes with smaller sample volumes will produce zero concentration in size ranges where the second probe with a larger sample volume provides a measurement. Thus, it is preferred to use the non-zero value.

If both probes have non-zero values for less than half of the crossover region:

Only the probe with more non-zero values is used for the entire crossover region. This prevents a few outliers from producing jumps in the distribution.

Uncertainty calculation:
Error bars can be calculated for two probes that overlap within a size range. A fixed uncertainty for that timestep is calculated within the overlap region, and applied across the full size range of both probes.

probe_endbins.overlap_cdp_holo = [15.65 17.61 19.57 21.53 23.49 25.45 27.41 29.37 31.34];
probe_endbins.overlap_holo_f2ds = [45. 55.   65.   75.   85.   95. 105. 125. 145. 175. 225. 275. 325. 400.];
probe_endbins.overlap_f2ds_hvps = [975. 1125.   1275. 1425.];

(*calculate_uncertainty.m*) Error is calculated as the mean of absolute difference between two probes within their overlap region: ONLY if over half the overlap region has two non-zero values (prevents a couple outliers from dominating the uncertainty values). Times where both probes equal zero are not used in the calculation (This would artificially reduce the error).

HVPS and other counting probes -> Uncertainty bar can be calculated based on square root of count number in bin. (Has not yet been implemented)

OPERATIONS DONE TO PREPARE DATA:

HOLODEC: Original HOLODEC Netcdf files are available in the CAESAR data archive, provided by Aaron Bansemer.

- Holodec data is rebinned to transition smoothly between CDP and F2DS bins.
  - See *holo_composite_rebin.py* in the *holodecRebin* folder.
  - Holodec mean diameter is used for particles with mean diameter < 45um and asprat>0.5, major diameter is used for mean diameter >=45um and any particle with asprat>0.5
- A 5 second boxcar average is applied to create the Holodec 1Hz PSD's, in order to increase particle statistics.
  - See *read_holo.m*

F2DS:

- F2DS distributions used here are provided by Sarah Woods and Aaron Bansemer, processed using SODA release with DoF compactness implemented, and Korolev resizing (PSC) for particles up to 250 um (same Korolev resizing as the archived data).
- F2DS bin edges [700. 800. 900. 1000. 1200. 1400. 1600. 1800. 2000.] are rebinned to [700. 825. 975. 1125.   1275. 1425. 1575. 1725. 1875.] to allow smooth transition with HVPS.
  - Concentrations are proportionally divided into the new bins.
  - See *read_f2ds.m*
- F2DS horizontal channel is used exclusively.

CDP:

- CDP Inboard is used exclusively.

- CDP bin edges [43.1 45.06 47.02] are rebinned to [43.1 45.00 47.02], to allow smooth transition with F2DS.
    - Concentrations are proportionally divided into the new bins.
    - See *read_cdp.m*

ROUND/ICE PARTICLE DISTRIBUTIONS:

(*PSD_phase.m*) The same algorithm as above is used to combine multiple probes into a best estimate particle size distribution. Differences in how the instrument data is prepared is documented below.

Round PSD:
The round particle distribution can be used as a rough estimate of liquid particles, based on the assumption that round particles are not ice. This is not always a safe assumption to make, so use caution. The round particle distribution uses only CDP, Round HOLODEC, and Round F2DS. HVPS is excluded as it is assumed large particles are not liquid. CDP cannot differentiate between round and non-round, and it is assumed that small particles are round.

Ice PSD:
The ice particle distribution contains particles that are not round. This can be used as a rough estimate of ice particles, based on the assumption that ice particles are not round. Use caution, as this is not always a safe assumption to make. The ice particle size distribution uses only Ice HOLODEC, Ice F2DS, and HVPS. CDP is excluded as it cannot differentiate between round and non-round.

Round/Ice HOLODEC:
(*read_holo.m*) HOLODEC particles are counted as "round" according to the following conditions:

- Greater than 0.5 circularity for particles of mean diameter less than 26um.
- Greater than 0.7 circularity for particles of mean diameter less than 46um.
- Greater than 0.8 circularity for particles of mean diameter greater than 46um.

If the above conditions are not met, the particles are counted as "ice".

Round F2DS:
Provided by Sarah Woods. A per-particle roundness threshold of 0.7 is used. "Ice" concentrations are obtained by subtracting the F2DS round concentration from the F2DS total concentration.